Full Set of Prompts Used in Radiology TAT Study

1. I am a machine learning engineer. You are a Radiologist. I am trying to find out why the time interval from the MRI or CT Scan Finished Date to Report Signed Date is too long? I know there are policy and regulatory issues at stake, but I need to find out if there is a correlation between the two. As output, produce a table with factors (rows) and a description plus ways to further investigate (columns)

2. Please tell me more about 'Patient Factors'. Expand the table you generated, and add reasons why these factors may increase the time to come up with a report? Condition: find and use medical factors, if possible.

3. Print out the entire set of prompts I entered so someone else can replicate this session.

4. Please tell me more about the following factors: Radiologist Workload, Modality Type and Complexity, Dictation and Transcription Delays, Unstructured Notes or Speech Recognition Errors, Bottlenecks in Worklists, Missed Alerts or Inadequate Notification, Regulatory/Sign-Off Thresholds. Expand the table you generated earlier, and add reasons why these factors may increase the time to come up with a report. Condition: find and use medical factors, if possible.

5. Produce individual expanded table for each factor just like you did for the patient factors.

6. I will need datasets to investigate these causes of long time to come up with reports for radiology scans. What kind of datasets would I need? What are the features they should contain? Produce a list of data sources that would satisfy these requirements, with a link to access them.

7. You are a radiologist and an expert researcher in the medical field of radiology. I am a data scientist. Perform an extensive literature review on why the time interval from the Scan Finished Date to Report Signed Date of radiology scans is too long. As a result, produce the following: a problem statement, the current situation (that is the average time taken from scan to report in hours), the operational workflow process involved from scan finished to report signed, A 5-Why analysis on why the problem exists. Quote relevant authentic sources (preferably research papers) to support your figures and numbers. If possible, focus on Canada or the USA. Canada preferably.

8. This should focus on X-Rays, MRIs and CT-Scans. Outpatient/Imaging centers should be included. The

review should include interventions or strategies that have been proposed to reduce turnaround time.

9. From the 5-Why analysis, Identify the root cause of long time interval from the Scan Finished Date to Report Signed Date of radiology scans.

10. Produce a table that shows the 5 why analysis with 2 columns (question, answer) and 5 rows.

11. Print out the entire set of prompts I entered so someone else can replicate this session.

12. You are a radiologist and an expert researcher in the medical field of radiology. I am a data scientist. I would like to present this problem in a TBP A3 form. Produce the following: 1. Clarify the problem (current situation, ideal situation, gap). This should be in terms of reporting TAT, 2. Breakdown the problem (a breakdown of the radiology workflow process with a distribution of the gap. This will help identify the point of occurrence), 3. Set the target, 4. The 5-Why root cause analysis that identifies the root cause by asking questions from the point of occurrence, 5. Problem statement for this problem. All outputs should have strong literature backing and citations to support numbers and facts. Do not hallucinate the figures. The numbers for the reporting TAT should be factual and accurate. An average value will be okay.

13. Can we focus on a specific number for TAT not a range also for the workflow breakdown.

14. Your breakdown of the workflow does not align with the gap.

15. Why did the numbers change from 5 hours to 26 hours.

16. I need the exact source that gives the 26.6 hours time.

17. Lets choose one baseline to maintain consistency: If you're modeling from a Canadian hospital/outpatient system, use 26.6 hours. If modeling from an American outpatient/imaging center, use 5 hours. I am modelling from a Canadian hospital/outpatient system.

18. The breakdown should sum up to the gap of 20.6 hours, not the total of 26.6 hours.

19. | **Stage** | **Time (hours)** | **% of Total Gap** | **Description** |... For this table, I need you to rewrite this but name the stages as actual stages in radiology workflow.... Not the delays or delay reason.

20. Can you create a parent group for the 5 stages. Group it into 2 major phases.

21. Give me a brief description of the five workflow stages.

22. Make it shorter.

23. Hughes et al., Improving Radiology Turnaround Times, JACR, 2021. CIHI 2023 Report on Diagnostic Imaging. Lee et al., Workflow Analysis in Radiology Reporting, BJR, 2020. Give these references in APA format.

24. I need to create a research paper on this topic based on the research already done. I have attached the jupyter notebook containing exploratory data analysis done on the radiology turnaround data and statistical testing done. As an output, produce a research paper that should include the following sections: Abstract, Introduction, Problem statement, Targets, Root cause analysis, Objectives, Methodology, References. The paper may contain additional sections to present: The Implementation of the countermeasure, The results of experiments and tests of said application, Challenges, roadblocks, and limitations of your research, Next steps (in a possible future of the project), Add citations and references all throughout the paper (In APA referencing format).

25. I need this in a research paper format, not bullet points. You need to act as a researcher drafting a research paper.

26. I need this to be more detailed. You are a very detailed research radiologist. Also, provide at least 15 valid references.

27. Export as a Word document.

28. Print out all my prompts so that this result can be replicated.

29. I want it in a pdf document.